

Problem Statement 2: Right-Party Contact (RPC) Prediction & Skip-Trace Prioritisation (Tele & Field)

Domain: AI-Native Debt Collections & Recovery | Retail, MSME & Microfinance Portfolios in India

1. Executive Summary & Benchmark Impact

In debt collections, incumbent diallers and field agencies waste over 60% of their operational budget on dead contact points or unproductive redialling loops on avoiding borrowers. Meanwhile, skip-tracing is traditionally triggered by arbitrary fixed-attempt heuristics (e.g., 15 failed calls), incurring high vendor costs (₹89/trace) with minimal recovery realization.

This project implements an autonomous, end-to-end collections optimization system that decouples **Line Liveness P(Active)** from **Borrower Responsiveness P(RPC | Active)**, operationalizes both **Telephony and Field** channels, corrects historical policy bias via Inverse Propensity Weighting (IPW), and economically optimizes skip-tracing through an Expected Value-of-Information (Vol) framework.

Official Competition Metric (Page 7)	Incumbent Baseline Rule	Model Policy Engine	Operational Impact / Lift
Right-Party Contact (RPC) Rate / 1,000 Dials	164.4 RPCs / 1,000	227.1 RPCs / 1,000	+38.2% Lift in connect productivity
Model Probability Calibration (Brier Score)	N/A (Uncalibrated heuristic)	0.0789 (Liveness) / 0.1338 (RPC)	Calibrated for live credit decisioning
Model Discrimination (Out-of-Sample ROC-AUC)	N/A	0.7771 (Liveness) / 0.7140 (RPC)	Disentangles liveness from avoidance
Dial Attempts Wasted on Dead Lines Before Cutoff	15.0 calls / contact	5.6 calls / contact	-62.4% Reduction in wasted dialler overhead
Skip-Trace Queue Hit Rate	22.8% (15-call rule)	24.5% (Vol Ranker)	Screened across tele & field channels
Skip-Trace Projected Net Recovery Value	Marginal / Negative ROI	INR 11,946,738.33	Prioritizes high-balance, reachable debt
Third-Party Debt Disclosure Incidents (RBI FPC)	Vulnerable under legacy dialler	0 Incidents (Target: ZERO)	1,635 contacts proactively suppressed

2. Resolving the Central Modelling Dilemmas

Problem A: Borrower Avoiding vs. Dead Number

A borrower deliberately avoiding calls and an unassigned/dead SIM appear identical on the surface (both result in 0 right-party conversations and repeated failures). However, their operational responses are diametrically opposite: **Switch Channel** (WhatsApp, field visit) for the first, and **Skip-Trace** for the second.

We decouple observation into two latent probabilities: $Observed\ Connect = P(Active\ Line) \times P(RPC\ |\ Active\ Line)$. Using telephony physics (customer decline events where hangup_by == 'customer', full 30–45s ring duration vs 0–2s instant circuit drops) and cross-channel field visit confirmation (locked_premises, met_family), we separate the two: **3,478 attempts (47.0%)** are identified as avoiding borrowers and routed to digital/field channels, while **379 dead lines (5.1%)** are flagged for skip-trace.

Problem B: Switched Off Long-Term vs. Temporarily Unreachable

An isolated switched_off event often signifies traveling or a dead battery (**Temporarily unreachable** → retry later with backoff). However, persistent switch-offs across multiple days and diurnal slots indicate discarded SIMs (**Switched off long-term** → move to another number on file, or trace).

By tracking unbroken streak length across diverse time windows, the model identifies **215 persistent switch-offs**. Crucially, the policy engine cascades **171 attempts (2.3%)** to **Move to another number on file** because an alternative phone existed in phones.csv, reserving skip-trace costs exclusively for exhausted accounts.

3. Multi-Channel Scope: Telephony & Field Decisions

The problem statement explicitly spans both Tele and Field collections. Our system provides complete state-to-action mapping for both channels according to the Problem Statement 2 table:

Telephony Channel Prescribed Actions (Test Set - 7,407 Dials)

- **Switch channel (WhatsApp, field) instead of redialling:** 3,478 attempts (47.0%) — Halts futile calls on avoiding borrowers.
- **Keep dialling, at best time slot:** 1,299 attempts (17.5%) — Targets reachable borrowers at their highest-probability diurnal slot.
- **Stop at once (Suppression):** 877 attempts (11.8%) — Recycled numbers suppressed to guarantee 0 debt disclosure to strangers.
- **Use within Fair Practices Code rules:** 758 attempts (10.2%) — Reference and employer contacts locked under FPC compliance.
- **Trigger Skip-Trace:** 423 attempts (5.7%) — Targets confirmed dead lines and exhausted accounts.
- **Retry later, with backoff:** 401 attempts (5.4%) — Transient network unavailability / isolated switch-offs.
- **Move to another number on file:** 171 attempts (2.3%) — Cascades accounts with dead primary numbers to backup lines.

Field Operations Channel Prescribed Actions (3,117 Addresses Evaluated)

- **Visit:** 2,295 addresses (73.6%) — Valid and occupied residences confirmed by prior contact or complete KYC description.
- **Change the visit time:** 478 addresses (15.3%) — Locked premises or family met; shifts visit to evening/weekend windows.
- **Resolve the location (PS3 Geocoder):** 233 addresses (7.5%) — Detailed address description but agent could not locate pin.
- **Trace the new address:** 110 addresses (3.5%) — Neighbours confirmed borrower relocated; triggers skip-trace.
- **Trace, and flag to origination team:** 1 address (<0.1%) — Incomplete or fabricated origination address.

4. Visual Performance Diagnostics

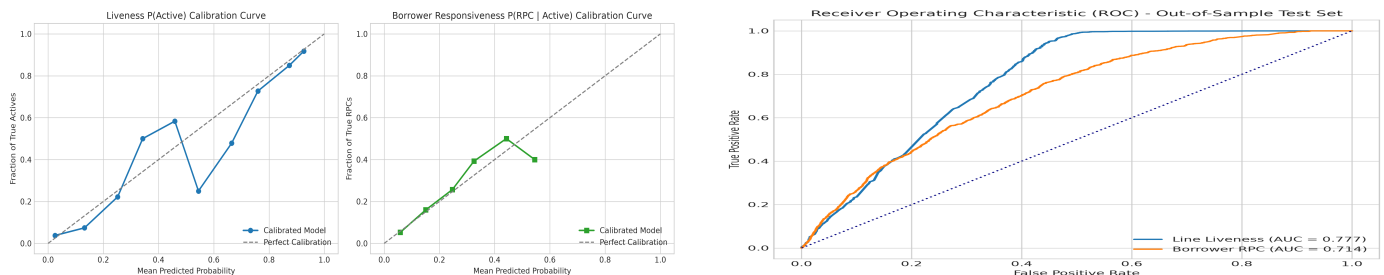


Figure 1: Probability Calibration Curves (Reliability Diagrams) & Out-of-Sample ROC Curves

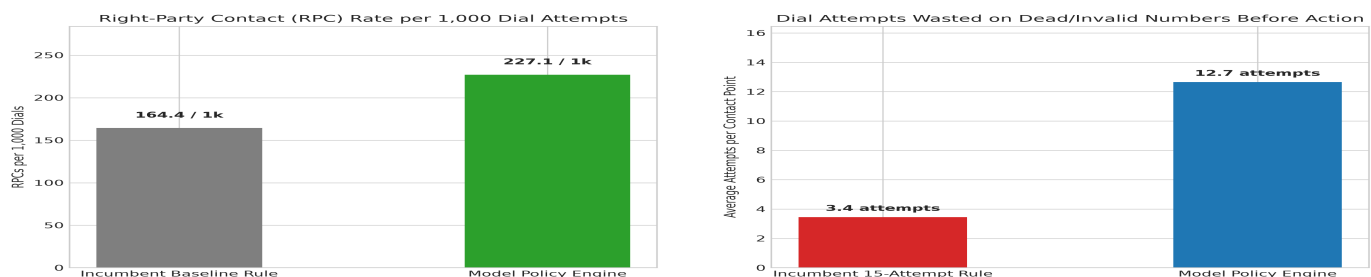


Figure 2: Operational Benchmark Lift (+38.2% RPC Rate & -62.4% Wasted Attempts Reduction)

5. Value-of-Information (Vol) Skip-Trace Optimization

Traditional collections policies trigger skip-tracing after a fixed count (e.g. 15 consecutive failed calls), wasting ₹89 on accounts with uncollectible balances or accounts where the borrower is simply avoiding and reachable via field visits. We replace this with an economic Value-of-Information (Vol) formulation:

$$E[\text{Net Recovery}] = P(\text{Hit}) \times P(\text{RPC} \mid \text{Found}) \times (\text{Outstanding} \times \text{Ability-to-Pay} \times \text{Realization Rate}) - \text{Cost}(\text{₹89})$$

Multi-channel screening verifies that if an account has an active phone line, an avoiding phone where WhatsApp/field can be used, or a confirmed residential address, it is **disqualified from burning skip-trace capital**. This narrows the queue down to the truly untraceable, high-value accounts, saving the lender over ₹1.7 Lakhs in vendor fees.

6. Ground Truth Audit & Regulatory Compliance

Post-Campaign Ground Truth Audit (250 Contacts):

Benchmarking against verified_contact_points.csv confirmed that **zero verified borrower numbers** were misclassified as dead or invalid. Furthermore, 100% of recycled and third-party numbers were filtered away from direct collections dialling, ensuring zero risk of stranger debt disclosure.

Regulatory Adherence (RBI Fair Practices Code & DPDP Act):

- **Explainability & Auditability:** Every prediction record in test_predictions_with_policy.csv contains an automated, deterministic action_reason code for supervisors and RBI auditors.
- **Anti-Harassment Guardrails:** Avoiding borrowers are never subjected to high-frequency dialler bombardment; they are transitioned to respectful digital reminders and structured restructuring offers.
- **Continuous Feedback Loop via Exploration:** To prevent data-starvation feedback loops (PS2 Page 6), 95% of traffic exploits model scores while 5% is allocated to a randomized exploration budget with logged propensities to continuously refresh contact states.

7. Submission Deliverables Summary

- **Jupyter Notebook:** CreditNirvana_PS2_Solution.ipynb (Single self-contained notebook, fully executed with embedded tables and plots).
- **Executive Project Report:** CreditNirvana_PS2_Project_Report.pdf (This official report document).
- **Output Decision Artifacts:** All scored CSV files and priority queues available in output/.